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Increase in antimicrobial resistance in *Escherichia coli* in food animals between 1980 and 2018 assessed using genomes from public databases

João Pires [1]*, Jana S. Huisman [2,3], Sebastian Bonhoeffer [2] and Thomas P. Van Boeckel [1,4]

[1]Institute for Environmental Decisions, ETH Zurich, Zurich, Switzerland; [2]Institute of Integrative Biology, ETH Zurich, Zurich, Switzerland; [3]Swiss Institute of Bioinformatics, Lausanne, Switzerland; [4]Center for Disease Dynamics, Economics & Policy, New Delhi, India

*Corresponding author. E-mail: joao.pires@env.ethz.ch

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Background: Next-generation sequencing has considerably increased the number of genomes available in the public domain. However, efforts to use these genomes for surveillance of antimicrobial resistance have thus far been limited and geographically heterogeneous. We inferred global resistance trends in *Escherichia coli* in food animals using genomes from public databases.

Methods: We retrieved 7632 *E. coli* genomes from public databases (NCBI, PATRIC and EnteroBase) and screened for antimicrobial resistance genes (ARGs) using ResFinder. Selection bias towards resistance, virulence or specific strains was accounted for by screening BioProject descriptions. Temporal trends for MDR, resistance to antimicrobial classes and ARG prevalence were inferred using generalized linear models for all genomes, including those not subjected to selection bias.

Results: MDR increased by 1.6 times between 1980 and 2018, as genomes carried, on average, ARGs conferring resistance to 2.65 antimicrobials in swine, 2.22 in poultry and 1.58 in bovines. Highest resistance levels were observed for tetracyclines (42.2%–69.1%), penicillins (19.4%–47.5%) and streptomycin (28.6%–56.6%). Resistance trends were consistent after accounting for selection bias, although lower mean absolute resistance estimates were associated with genomes not subjected to selection bias (difference of 3.16%±3.58% across years, hosts and antimicrobial classes). We observed an increase in extended-spectrum cephalosporin ARG *bla*$_{\text{CMY-2}}$ and a progressive substitution of *tetB* by *tetA*. Estimates of resistance prevalence inferred from genomes in the public domain were in good agreement with reports from systematic phenotypic surveillance.

Conclusions: Our analysis illustrates the potential of using the growing volume of genomes in public databases to track AMR trends globally.

Introduction

Antimicrobial resistance (AMR) is a growing threat fuelled by the overuse of antimicrobials in humans and animals.[1,2] Globally, two-thirds of antimicrobials are used in animals raised for food.[3,4] This creates substantial selective pressure that facilitates the emergence and spread of antimicrobial resistance genes (ARGs) in food animals.[5]

Next-generation sequencing (NGS) helped identify genetic determinants of resistance in bacteria.[6] However, isolates associated with animals have received comparatively less attention than isolates found in humans.[7,8] Of the 108 392 *Escherichia coli* genomes in the Pathogen Detection database thus far,[9] only 7626 (7.03%) were recovered from food animals compared with 73 340 (67.7%) from humans. Given the ubiquity of *E. coli* in humans and animals, and its increasing resistance rates,[10] monitoring its prevalence in animals would help fill an important knowledge gap. NGS also led to a considerable increase in the number of genomes deposited in public repositories.[6,11] This growing number of 'public genomes' complements existing surveillance systems monitoring AMR trends based on phenotypic testing.[12–14] Thus far, phenotypic methods have been used for systematic surveillance of AMR in high-income countries,[13,15] as well as in epidemiological studies documenting resistance in low- and middle-income countries (LMICs).[16] However, these techniques cannot identify individual ARGs or their association with mobile genetic elements (MGEs) or specific strains, although this has shown to be useful to guide interventions.[6]

Genomic surveillance presents several advantages over phenotypic assays.[6] First, it allows us to distinguish whether resistance was acquired via mutations or acquired ARGs.[6] Second, it enables the performance of *in silico* molecular typing of bacterial lineages and plasmids associated with ARGs.[17,18] This facilitates tracking AMR between host species,[19–21] such as *mcr* genes in pigs and humans,[22–25] as well as enabling backward comparability with past surveys.[26,27]

Despite these advantages, inferring the resistance levels from ARGs retrieved from NGS data presents challenges.[28] In particular, NGS remains expensive and assembly quality checks are essential to ensure comparability across NGS platforms and to extract epidemiologically-relevant information from public genomes. Additionally, virulent/resistant genomes and specific bacterial strains may be over-represented in public databases.[29–31] The impact of this potential sampling bias must be quantified for inferring resistance trends. Finally, the lack of standardized metadata might result in the exclusion of relevant entries from data analysis.[29,30,32]

In this study, we report on trends in the prevalence of ARGs in poultry, bovines and swine using public genomes. We compare resistance trends inferred from genomic data with trends reported by systematic [National Antimicrobial Resistance Monitoring System for Enteric Bacteria (NARMS) and European Food Safety Agency (EFSA)] and passive (resistancebank.org) surveillance systems.

Methods

Harmonization of genomic data

We searched for genome assemblies (hereafter referred to as 'genomes') of *E. coli* deposited until 31 December 2018 in three public databases: the Nucleotide Database of the National Center for Biotechnology Information (NCBI),[33] EnteroBase[34] and the Pathosystems Resource Integration Center (PATRIC).[35] We filtered metadata for entries related to food-producing animals using text matching. A harmonized dataset of unique entries was generated [n = 7632; Figure S1 and Supplementary Information (available as Supplementary data at *JAC* Online)].

Identification of selection bias

We used the BioProject metadata associated with each genome to identify if isolates had been preferentially selected for sequencing based on virulence, resistance or specific strains (e.g. serotypes, sequence types and pathotypes). Metadata were queried using Entrez (Dataset S1).[36] For BioProjects with insufficient metadata, we consulted associated publications to identify potential selection criteria. We classified the BioProject sampling scheme ('Study Type') into 'Surveillance', 'Cross-Sectional', 'Strain Survey', 'Investigation' and 'Unknown'. Thereafter, we assigned the values ('Yes', 'No' or 'Unknown') to the following criteria: 'Resistance Selection', 'Virulence Selection' and 'Strain Selection'. All entries with the value 'Unknown' for at least one selection criterion were assigned to the 'Selected' group of genomes. We created two datasets with genomes isolated between 1980 and 2018 from bovines, poultry and swine hosts. First, a dataset containing genomes exclusively from 'Surveillance', 'Cross-Sectional', 'Strain Survey' and with 'No' in all selection categories (n = 4201; hereafter referred to as 'non-selected') and a dataset containing all genomes irrespective of potential selection criteria (n = 6763; hereafter referred to as 'complete').

Bioinformatic analysis

The phylogenetic group (PhG) was determined *in silico* using ClermonTyping (version 1.0.0)[37] and MLST performed with mlst (version 2.16.2).[38] PhGs, sequence types and clonal complexes for EnteroBase genomes were retrieved from the metadata tables. *E. coli sensu strictu* genomes (PhGs A, B1, B2, C, D, E and F; Figures S2 and S3) were kept for further analysis.[27] ARGs were screened using ABRicate (version 0.8.13)[39] with the ResFinder database (version of 22 April 2019).[40] Matches with an identity and coverage above 80% were kept and assigned a resistance phenotype (Supplementary Information).

Trends in resistance

All ARGs were considered functional and thus isolates were classified as microbiological resistant to predicted resistance phenotypes (Supplementary Information).[41]

Three temporal trends were tracked: MDR, resistance prevalence by antimicrobial class and ARG prevalence. Genomes were assigned an MDR score: the number of antimicrobial classes a genome is predicted to confer resistance to (by carrying ARGs; Supplementary Information). The temporal trend of the MDR score was analysed using a negative binomial regression model and prevalence of resistance to individual antimicrobials and ARG prevalence were assessed using logistic regressions with hosts and years as independent variables. All models were trained and cross-validated using 10-fold cross-validation with the caret package.[42] Observations were weighted by the countries' Animal Population Correction Unit (PCU) and proportion of genomes contributed by a given country per year (Dataset S2 and Supplementary Information).

For the non-selected dataset, we used Monte Carlo simulations to build a dataset of equal size to the 'complete' dataset to ensure comparability of statistical analysis for the antimicrobial resistance and ARG prevalence analysis (equal sample sizes; n = 6763). Concretely, we supplemented the 'non-selected' dataset (n = 4201) with 2562 randomly selected additional genomes using resampling without replacement. The procedure was repeated 100 times to average the effect of resampling. Temporal trends were considered significant if the temporal coefficient was statistically significant in at least 50% of the simulations.

Results

Genomic data

We identified 7632 unique genomes: 5617 (Figure S1 and Table S1) from EnteroBase (73.6%), 2010 from PATRIC (26.3%) and 5 from NCBI (0.07%). Genomes originated predominantly from the USA (n = 4841, 63.4%), followed by Europe (n = 1320, 17.3%), Asia (n = 1006, 13.2%), Africa (n = 129, 1.7%), Central and South America (127, 1.7%) and Oceania (n = 88, 1.2%). Most genomes were recovered from bovines (n = 3956, 51.8%), followed by poultry (n = 2021, 26.5%) and swine (n = 1250, 16.4%). A total of 24 719 ARGs were identified (167 unique ARGs; Table S2).

Accounting for potential selection bias

Among the 7632 genomes recovered from public databases, most (62.51%, n = 4771) had not been selected for sequencing based on resistance/virulence or strain characteristics. Among selected genomes (n = 2861), the leading selection criterion was strain characteristics (n = 1425, 46.73%), followed by resistance (n = 978, 34.18%), virulence (n = 73, 2.55%) or a combination of factors (n = 11, 0.38%). The percentage of genomes subject to selection

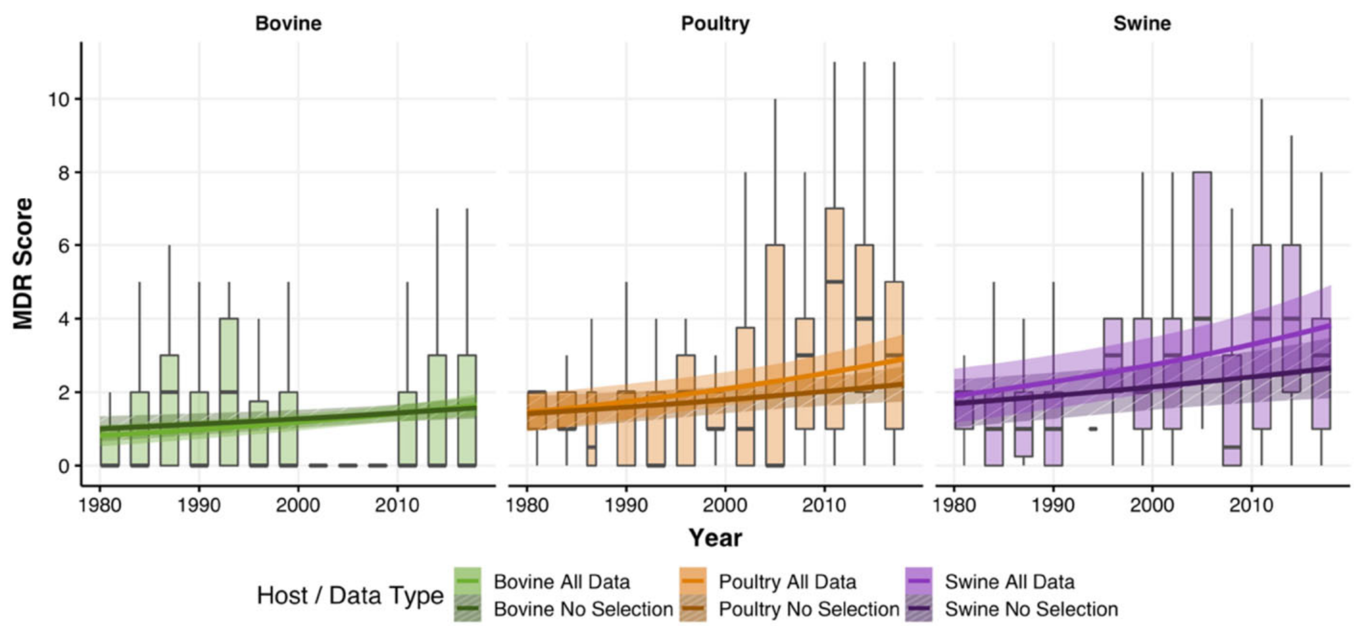

Figure 1. Temporal trends of MDR in food animals. Number of antimicrobial classes an *E. coli* genome isolated from food animals carries resistance to (MDR score). Boxplots show the distribution of the MDR score in the total dataset. The solid line represents a significant temporal trend for MDR scores obtained from the negative binomial regression and the shading the 95% CI. This figure appears in colour in the online version of *JAC* and in black and white in the print version of *JAC*.

was not associated with an increasing/decreasing trend since 2004 (year of introduction of NGS; Figure S4).

Global trends in resistance

We observed an increase in the MDR score over time (Figure 1). In the non-selected dataset in 1980, genomes carried, on average, genes conferring resistance to 1.69 antimicrobial classes in swine, 1.41 in poultry and 1 in bovines (Table S3). By 2018, rates increased 1.6 times: genomes carried, on average, genes conferring resistance to 2.65 antimicrobial classes in swine, 2.22 in poultry and 1.58 in bovines. Using the complete dataset, rates doubled from 1980 to 2018: from 1.90 to 3.82 in swine, 1.45 to 2.91 in poultry and 0.82 to 1.65 in bovines. The 95% CIs of estimates of MDR scores inferred from the 'non-selected' and the 'complete' datasets overlapped between 1980 and 2018 (Table S4).

The countries with at least five genomes with the highest median MDR score across all years were China (8), Brazil, Hungary and Malaysia (6 each) and Czechia, Germany, France and Italy (5 each; Figure 2). The USA and Belgium had a median MDR score of 1, while several other countries had a median score of 0, including Argentina, Canada, Chile, Japan and Sweden.

Antimicrobial resistance trends

Between 1980 and 2018, our analysis showed significant increasing temporal trends for penicillins, first- and second-generation cephalosporins and extended-spectrum cephalosporins in the non-selected dataset (Figure 3a, Figures S5 to S20 and Table S5). We identified additional increasing temporal trends for phenicols, penicillins in combination with β-lactamase inhibitors, cephamycins and monobactams in all simulations and for trimethoprim in 58% of simulations in the Monte Carlo approach (Figure 3b and Table S6). Using the complete dataset, we identified significant temporal trends in the same antimicrobial classes as in the Monte Carlo approach, but also for gentamicin (Figures S21 to S37 and Table S7).

In the non-selected dataset in 2018, high resistance levels (>40%) in swine were observed for tetracyclines (69.11%, 95% CI = 57.19%–81.04%), streptomycin (47.58%, 34.43%–60.72%) and penicillins (47.53%, 33.73%–61.32%). The remaining antimicrobials had resistance levels below 40%, including sulphonamides (38.02%, 25.27%–50.77%), trimethoprim (17.16%, 6.15%–28.17%), phenicols (13.89%, 3.86%–23.92%), extended-spectrum cephalosporins (11.87%, 2.06%–21.67%) and gentamicin (7.87%, 0.5%–15.23%).

In poultry, high estimates were identified for tetracyclines (59.47%, 50.73%–68.21%), streptomycin (54.59%, 45.67%–63.5%) and sulphonamides (42.04%, 33.19%–50.89%). Lower resistance levels were observed for penicillins (36.84%, 27.77%–45.91%), gentamicin (34.43%, 24.69%–44.17%), early-generation cephalosporins (33.92%, 25.04%–42.8%) and extended-spectrum cephalosporins (10.84%, 4.42%–17.25%).

In bovines in 2018, only tetracyclines (42.24%, 35.18%–49.3%), streptomycin (31.21%, 24.68%–37.73%) and sulphonamides (29.86%, 23.41%–36.31%) had prevalences above 30%. The remaining antimicrobials had resistance levels below 20%, including penicillins (19.43%, 13.68%–25.19%), phenicols (18.7%, 12.48%–24.93%) and extended-spectrum β-lactams (12.52%, 7.13%–17.91%).

The mean absolute difference in the prevalence obtained from the non-selected dataset compared with the simulated datasets was 0.019%±0.022% in bovines, 0.031%±0.041% in poultry and 0.053%±0.097% in swine. Compared with the complete dataset,

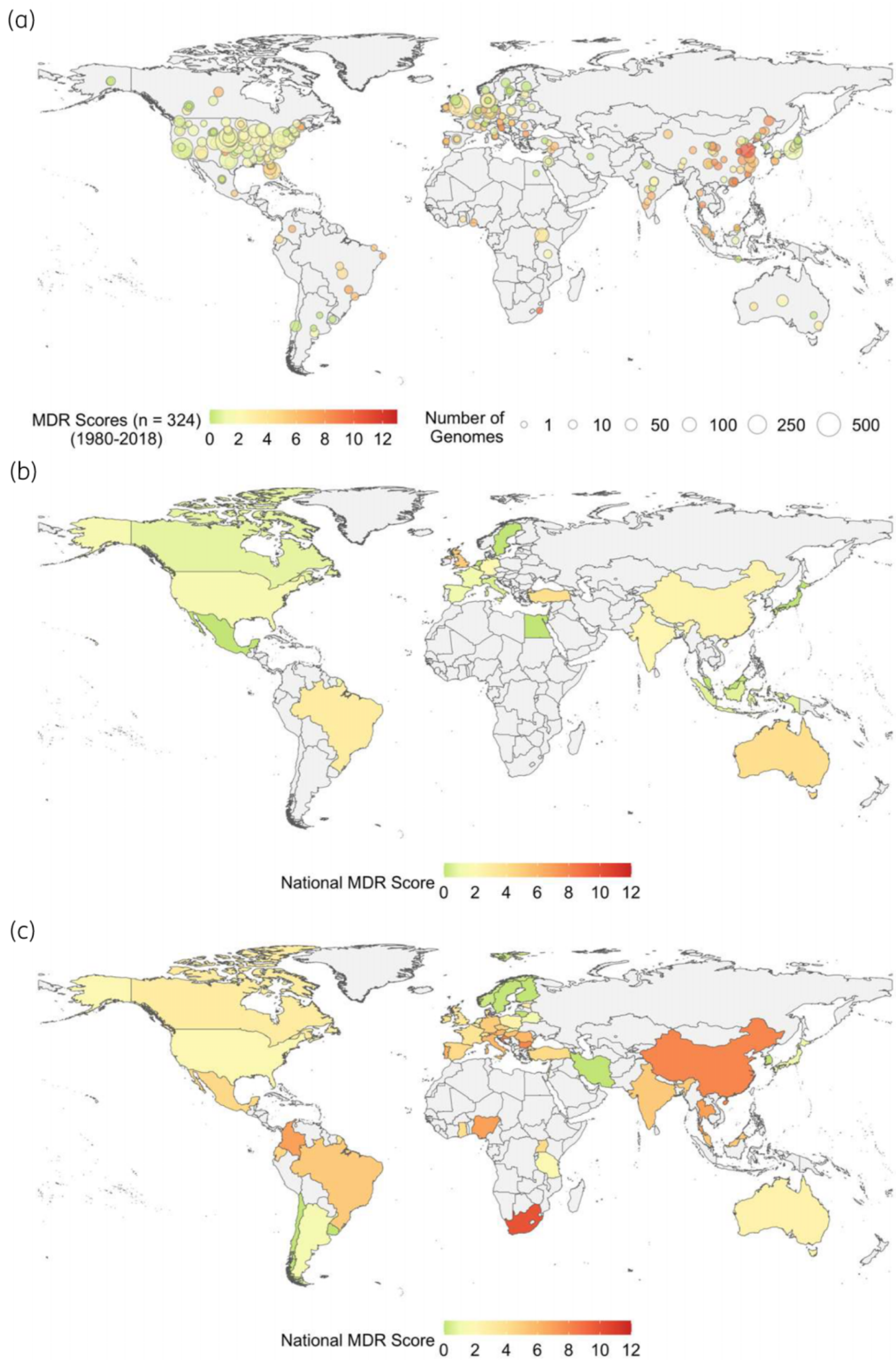

Figure 2. Global distribution of the average number of resistant antimicrobial classes per *E. coli* genome (MDR score) in food animals. (a) Distribution of the MDR score aggregated at unique geographical coordinates ($n = 324$) from 1980 to 2018. (b) National MDR score values from 2000 to 2018. (c) National MDR score values from 1980 to 1999. This figure appears in colour in the online version of *JAC* and in black and white in the print version of *JAC*.

the mean absolute difference in prevalence across antimicrobial classes was 1.29%±1.62% in bovines, 2.36%±1.72% in poultry and 5.82%±4.67% in swine.

For the time-series of resistance to individual antimicrobials, we estimated higher prevalence for the complete dataset compared with the non-selected dataset (Supplementary Information).

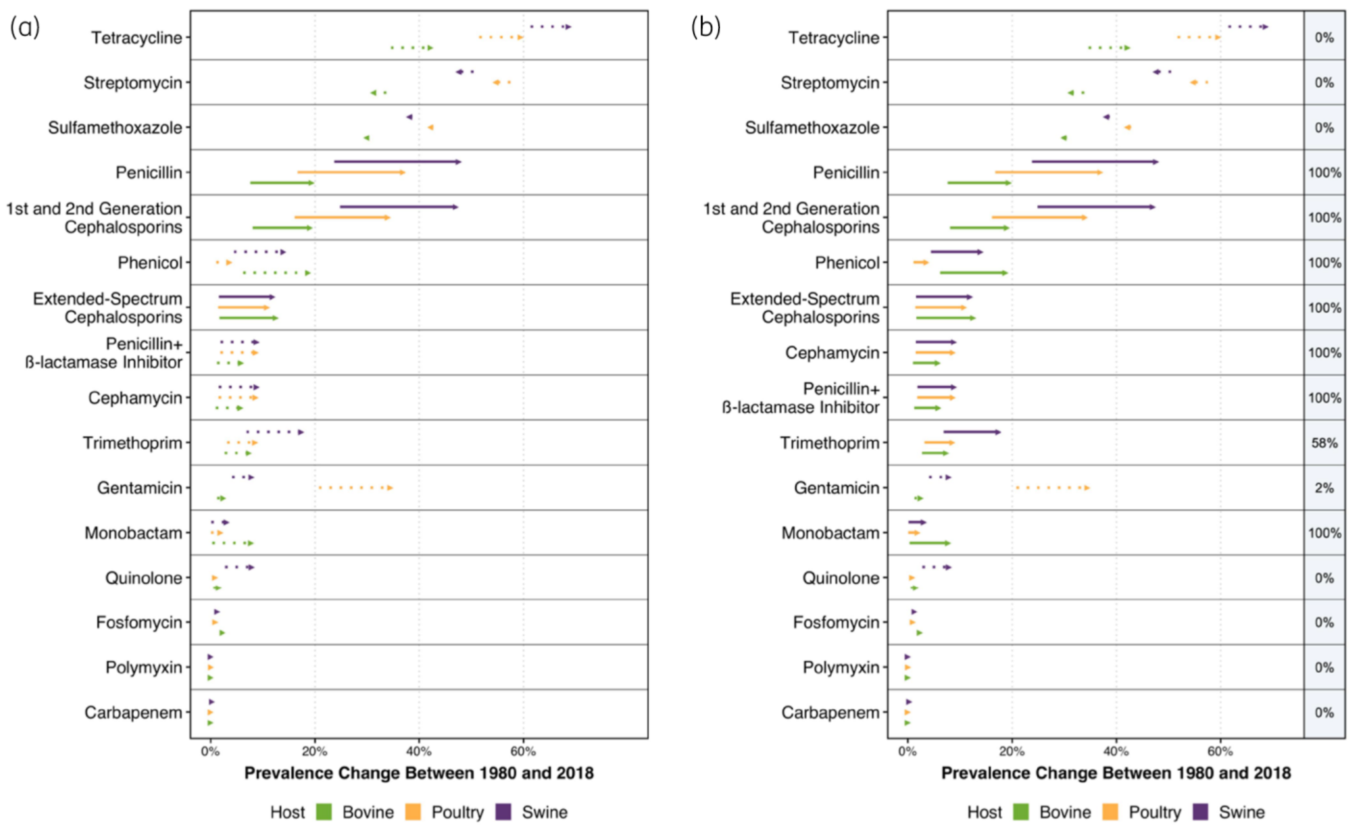

Figure 3. Change in resistance prevalence inferred from ARGs that confer resistance for 16 antimicrobial classes between 1980 and 2018 for the complete dataset. (a) Non-selected dataset. Solid lines indicate a statistically significant temporal trend (5% level) and dotted lines the absence of a significant temporal trend. (b) Monte Carlo simulations. Percentages indicate the percentage of statistically significant coefficients across all bootstraps. Solid lines indicate a statistically significant temporal trend in at least 50% of simulations and dotted lines the absence of a significant temporal trend. This figure appears in colour in the online version of *JAC* and in black and white in the print version of *JAC*.

The mean absolute difference was below 5% for all antimicrobials in poultry and bovines, except for tetracyclines in bovines (5.63%). For swine, this difference was higher than 5% for 7 out of 16 antimicrobial classes: gentamicin (11.95%), phenicols (11.38%), trimethoprim (10.86%), quinolones (10.12%), sulfamethoxazole (10.1%), streptomycin (7.7%) and penicillins (6.45%).

Prevalence comparison with international surveillance

We compared the prevalence of resistance inferred from ARGs with phenotypic surveillance data available for the USA,[15] Europe[43,44] and LMICs (resistancebank.org)[16] (Table S8). Our predictions using both datasets were consistent with those reported by these surveillance systems.[15,16,43,44] Our predictions for extended-spectrum cephalosporins (14.8%–21.2% and 10.84%–12.52% for the complete and non-selected datasets, respectively) were lower than those reported in LMICs, but higher than the USA (0.4%-6.3%) and Europe (0%–4%). For gentamicin, resistance predicted from genomes in poultry (38.5% and 34.43% for the complete and non-selected datasets, respectively) was higher than that reported in EFSA (4.9%–7.1%) and NARMS (22.3%–23.5%). For swine, gentamicin resistance was predicted to be higher in swine using the complete dataset (25.2%) compared with data reported in NARMS (2.5%), EFSA (2.6%) or resistancebank.org (5.6%–34.6%). Finally, our predictions for polymyxin resistance in swine using the non-selected (0.01%) or the complete (9%) datasets were lower than in LMICs (15.9%–36.1%).

Antimicrobial resistance gene trends

We analysed prevalence trends of individual ARGs (Figure 4, Figures S38 to S45 and Tables S9 and S10). Among all hosts in the non-selected dataset, a consistent pattern was observed for tetracycline ARGs: a progressive substitution of *tetB* by *tetA* (Figure 4a). *tetA* increased from 7.7% to 40.0% in swine, from 7.9% to 40.7% in poultry and from 4.4% to 27% in bovines, whereas *tetB* decreased from 55.4% to 31.4% in swine, from 46.8% to 24.5% in poultry and from 31.7% to 14.6% in bovines. Increases were also observed for *floR* (from 0.4% to 4.2% in swine, from 0.1% to 1.4% in poultry and from 2.1% to 17.6% in bovines). Both *aph-(6)-Ic* (from 10.8% to 0.3% in swine, from 14.5% to 0.5% in poultry and from 2.3% to 0.1% in bovines) and *aph(3')-IIa* (from 11.1% to 1% in swine, from 12% to 1% in poultry and from 3.6% to 0.3% in bovines) decreased in all hosts. Host-specific dynamics were observed in poultry: an increase in *bla*$_{\text{TEM-1B}}$ (from 9.8% to 20.1%) and decreases in both *aph(6)-Id* (from 53.3% to 20.6%) and *aph(3'')-Ib* (from 51.9% to 24.4%).

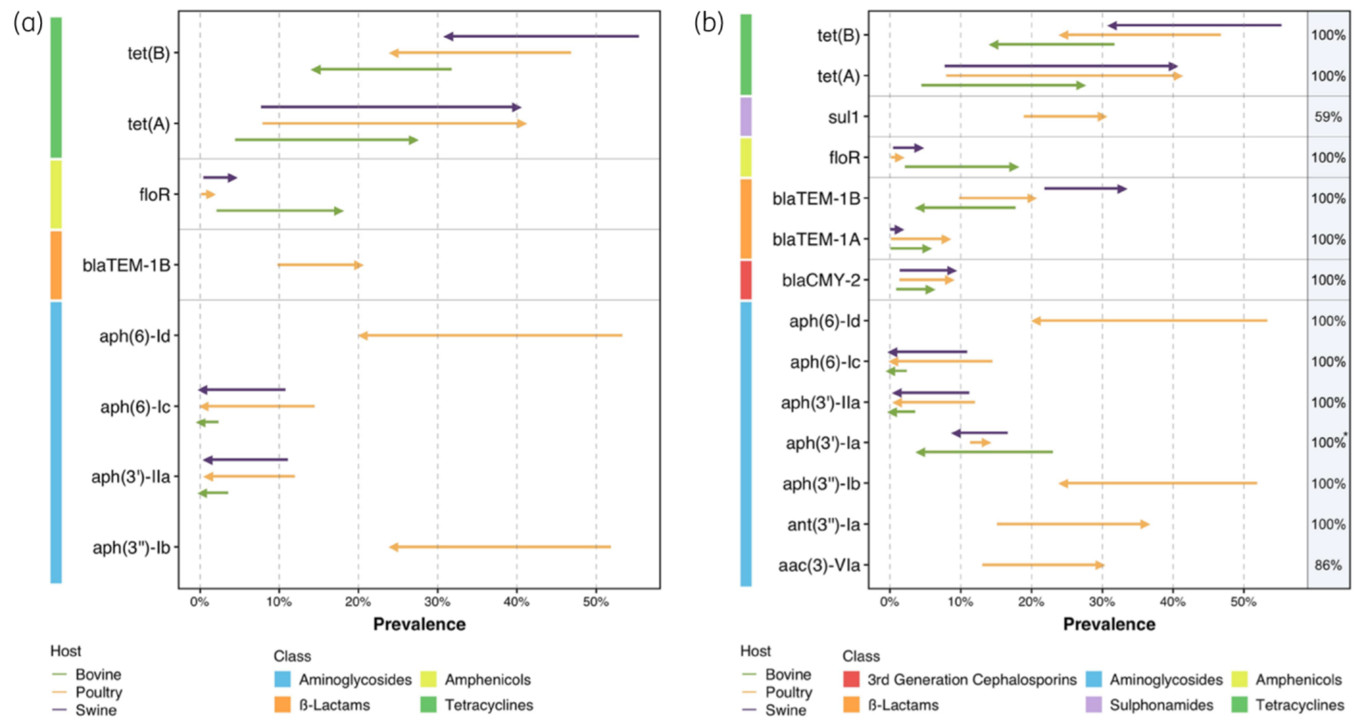

Figure 4. Change in the prevalence of ARGs between 1980 and 2018. (a) Non-selected dataset. Only significant temporal trends are shown (5% level). (b) Monte Carlo simulations. Percentages indicate the percentage of statistically significant coefficients across all bootstraps. Only statistically significant temporal trends in at least 50% of simulations are shown. *The interaction between poultry and year was in 99% of simulations. This figure appears in colour in the online version of *JAC* and in black and white in the print version of *JAC*.

In the Monte Carlo simulations (Figure 4b), we identified additional increasing trends in $bla_{CMY\text{-}2}$ (from 1.3% to 8.2% in poultry, from 0.9% to 5.7% in bovines and from 1.4% to 8.8% in swine) and $bla_{TEM\text{-}1A}$ (from 0.02% to 1.4% in swine, from 0.1% to 8% in poultry and from 0.1% to 5.3% in bovines). *aph(3')-Ia* decreased in bovines (from 23% to 4.3%) and swine (from 16.6% to 13.7%), but increased in poultry (from 11.3% to 13.7%). $bla_{TEM\text{-}1B}$ also increased in swine (from 21.8% to 32.9%), but decreased in bovines (from 17.7% to 4.2%). Poultry-specific dynamics were observed for *ant(3")-Ia* (from 15.1% to 36.1%), *aac(3)-VIa* (86% of the simulations, from 13% to 29.8%) and *sul1* (59% of the simulations, from 18.9% to 30.1%).

ARG trends inferred from the complete dataset were comparable to the Monte Carlo simulations (Figures S46 to S55 and Table S11). This included the dynamics observed for *tetA, tetB, floR,* $bla_{TEM\text{-}1B}$, $bla_{CMY\text{-}2}$, *aph(6)-Ic, aph(3')-Ia* and *aph(3')-IIa* in all hosts, and the *aph(6)-Id* decrease in poultry.

Discussion

The increase in the prevalence of ARGs and their accumulation in MDR organisms is a growing concern for animal health and human health.[16] Thus far, AMR surveillance has predominantly relied on phenotypic testing.[12,14] Here, we used public information on genomic surveillance to document AMR trends in *E. coli* in food animals globally.

One of the challenges of using 'public genomes' is the potential over-representation of genomes based on strain, resistance or virulence characteristics.[45] We attempted to account for and quantify this potential bias by identifying genomes that were selected for these criteria based on their BioProject descriptions. We created two datasets ('non-selected' versus 'complete') and compared the trends in AMR inferred from each. Our mean resistance estimates were lower using the non-selected dataset, but within the 95% CI obtained by the two approaches (Figure 1).

According to the trends inferred from each dataset, MDR in food animals is rising globally. Between 1980 and 2018, MDR increased by a factor of 1.6 to 2 across hosts (Figure 1). This increase could be attributable to co-resistance to antimicrobials commonly used in agriculture such as tetracyclines, streptomycin and sulphonamides that were associated with high resistance rates (Figure 3 and Figures S56 to S59).[46] Perhaps of greater concern is the increase in resistance to critically important antimicrobials, including extended-spectrum β-lactams (Figure 3 and Figure S21). We identified comparatively lower MDR scores in bovines compared with other animals. This could be attributed to lower antimicrobial use compared with poultry and swine.[3,47]

Our findings are generally consistent with resistance trends reported from phenotypic international surveillance programmes.[15,16,43,44] For gentamicin and extended-spectrum cephalosporins, prevalence estimates obtained from public genomes were higher than those reported in international surveillance (Table S8). These differences can be difficult to explain given the different antimicrobial resistance prevalence reported in different countries. However, for these antimicrobials, resistance levels reported in the USA could be substantially underestimated because the breakpoint used for susceptibility testing by NARMS is 8-fold higher than that of EFSA for third-generation cephalosporins

(≥4 and ≥0.5 mg/L, respectively, for cefotaxime) and 4-fold higher for gentamicin (≥16 versus ≥4 mg/L, respectively).[13,48] Despite the lower prevalence estimates obtained in the non-selected dataset, these were still higher than those reported in international surveillance, indicating potential preferential sequencing of isolates for these resistance phenotypes. For polymyxin resistance, the lower estimates in our analysis compared with LMICs could be due to the small number of *mcr*-carrying genomes from Asian countries in public databases.[49,50]

To date, *E. coli* ARG prevalence has predominantly been associated with cross-sectional studies[10,51] and temporal resistance trends have been estimated using phenotypic data.[52] The combined analysis of non-selected and Monte Carlo simulations shows that some ARGs presented the same temporal trends in all hosts (Figure 4), suggesting that there may be a common gene pool shared by food animals.[20,53] The additional temporal trends identified by simulations also indicate that some genes might be influenced by local epidemiology.

We identified a pattern of progressive replacement of genes over time in which *tetA* has replaced *tetB* (Figure 4), which could not have been observed from phenotypic data alone. Furthermore, the increase in bla_{CMY-2} is concerning as this gene has been commonly identified in plasmids that might facilitate its dissemination in food animals.[54–56] Moreover, bla_{CMY-2}-carrying plasmids in humans are similar to those in food animals,[57] highlighting the need to perform studies quantifying the transmission and directionality of bla_{CMY-2}.

The increase observed for *floR* also highlights differences in antimicrobial usage in animals compared with humans, since this gene confers resistance to florfenicol, which is only approved in veterinary medicine in the EU and the USA.[58,59] Other dynamics of gene substitution were host-specific: decrease in bla_{TEM-1B} in bovines and aminoglycoside ARGs in poultry (Figure 4 and Figure S46). Animal-specific trends could suggest that either different bacterial populations and/or MGEs circulate in animal hosts or that different selection processes between animals lead to the increase in relative frequency of ARGs in different bacterial populations.[60]

Future directions

Our results can help identify areas for action. Our work shows that although selection bias exists, its impact can be quantified and inferences from genomic data can be made to track resistance. Increased data sharing and annotation of public repositories may help progressively reduce selection bias and improve AMR surveillance globally.[61] Limitations associated with using public genomes could be mitigated by a wider adoption of the FAIR principles: findable, accessible, interoperable and reusable.[62] This transition can be facilitated by researchers and database managers. Researchers are responsible for providing contextual metadata and should avoid bypassing mandatory fields by filling them with 'missing'. Currently, 47.2% of genomes in public databases lack an identifiable host.[9] Databases can help standardize metadata by matching terms to existing ontologies for genome submissions.[63] This would result in metadata available as a controlled vocabulary.[32,64]

International comparisons of AMR trends with genomic databases may be strengthened in the future, in particular between Europe and the USA. NARMS is already using NGS in surveillance[15,65] and EFSA anticipates that EU member states will adopt NGS by 2025.[13] This means that the number of surveillance-derived genomes will increase substantially and is an opportunity to establish analysis pipelines as described here. In contrast, LMICs are currently poorly represented in public databases and this precludes making robust inferences on ARG prevalence trends in these regions. The lack of genomic data from LMICs is particularly acute in South America, where several countries are major meat exporters.[66,67] Capacity building in these regions is urgent to better understand regional AMR trends.[68,69]

Our results raise new questions related to ARG persistence and spread. This is highlighted by the increase in bla_{CMY-2}, as extended-spectrum β-lactams are not approved for use as growth promoters in food animals in the USA and Europe.[59] The presence of this ARG in food animals could be linked to the use of these antimicrobials prior to their ban[59] or to co-selection, since isolates carrying ESBLs and/or AmpCs are often resistant to other antimicrobials.[59] On-farm monitoring of antimicrobial use and ARGs could help better understand this phenomenon. Moreover, the ARG substitution dynamics (e.g. *tet* genes) and animal-specific ARG trends (Figure 4) show the added value of using genomic data over phenotypic surveillance.[58] However, factors underlying these trends remain unknown. Changes in farming practices may favour certain *E. coli* strains over others,[70,71] which may contain different ARGs and MGEs.[17,71] Indeed, *tetA* and *tetB* have been associated with different insertion sequences, so the successful association of *tetA* with specific MGEs and/or strains compared with *tetB* is a potential hypothesis to explain these dynamics.[72,73]

Limitations

Our analysis comes with limitations. Firstly, 63.4% of the genomes come from the USA. To account for the over-representation of certain countries, we allocated each country a statistical weight proportional to the quantities of food animals expressed as the PCU.[3] Secondly, most genomes (69.7%) were collected between 2010 and 2018. Whilst regional assessment is currently only feasible for the USA (Supplementary Information and Figures S60 and S61), the increasing number of sequences could make similar assessments possible for other country/regions within years. Thirdly, we accounted for the potential bias in databases by screening the metadata available in the BioProject descriptions. Nonetheless, the lack of standardized metadata fields renders it challenging to identify: selection bias towards specific PhGs as it is not systematically reported; the sampling scheme used in different studies; and the reasons underlying the sequencing of specific isolates. Ensuring correct characterization of sampling bias could be facilitated by adopting dedicated standardized metadata fields. Fourthly, in the bioinformatic analysis we used assemblies instead of raw reads. While differences can be observed in the prediction of ARGs using reads,[74] the accuracy of such approaches is only ∼3% higher compared with assembly-based approaches.[74] The choice to use assemblies is in line with recommendations for open pathogen surveillance advocating databases to store ready-to-use assemblies to prevent repeating bioinformatic preprocessing.[61]

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Transparency declarations

None to declare.

Data and materials availability

All assemblies used in this study are available in public databases. All identifiers for traceability, including database of origin, database internal ID and BioSample (when applicable), are available in Table S1. All code used for this manuscript is available at Zenodo.[75]

Supplementary data

Supplementary data, including Figures S1 to S61, Datasets S1 and S2 and Tables S1 to S11, are available at *JAC* Online.

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